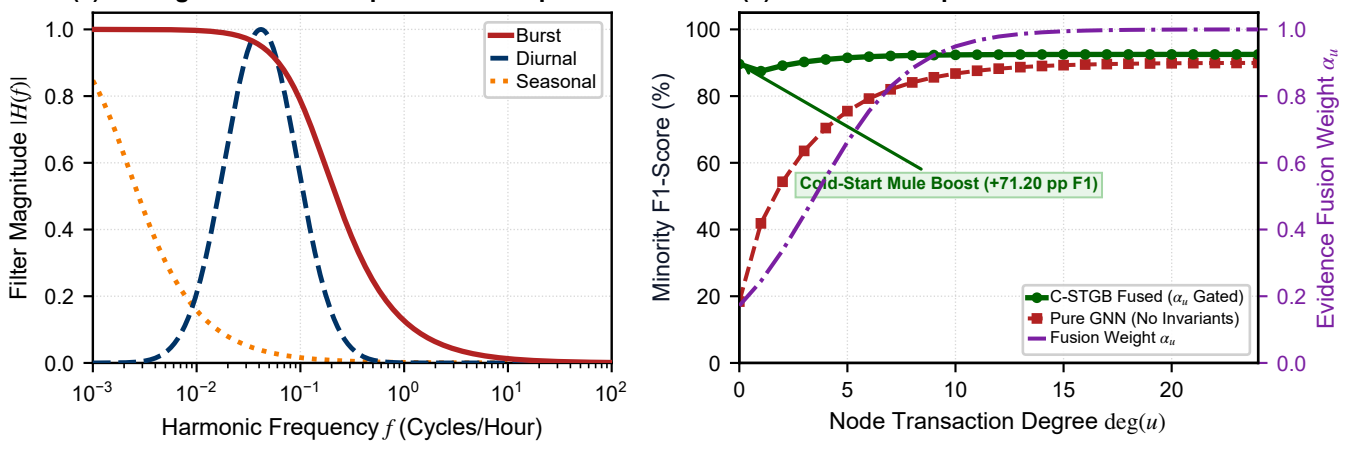

(a) Orthogonal Tri-Band Spectral Decomposition
Filter Magnitude $|H(f)|$
Harmonic Frequency f (Cycles/Hour)
Burst
Diurnal
Seasonal
(b) Evidence-Adaptive Cold-Start Resolution
Minority F1-Score (%)
Evidence Fusion Weight α_u
Node Transaction Degree $\deg(u)$
Cold-Start Mule Boost (+71.20 pp F1)
C-STGB Fused (α_u Gated)
Pure GNN (No Invariants)
Fusion Weight α_u